

Effective fake news video detection using domain knowledge and multimodal data fusion on youtube

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ABSTRACT

In the digital age, numerous videos are being actively produced and uploaded online. Simultaneously, fake news videos to attract public attention are also on the rise. Therefore, intensive research is being conducted to detect them. Validating video content is critical for all users as the public is exposed to various fake news videos. This study proposes ways to detect fake news videos effectively using domain knowledge and multimodal data fusion. We use domain knowledge to perform learning by reflecting the potential meaning of comments, helping us detecting fake news videos. We also use the linear combination to efficiently adjust the encoding rate for each characteristic of the video and effectively detect fake news videos. In particular, the domain knowledge improves the model performance by approximately 3% for all test datasets. Consequently, we achieve an F1-score of 0.93, which is higher than those of other comparison models in all the test datasets.

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1. Introduction

Since its introduction in 2005, YouTube has grown from a site that focused on amateur videos to one that distributes original content. The public has watched various videos in their daily lives through YouTube and other video media. In the digital age, the public, who used to be one-sided consumers, can easily produce and upload videos online, implying that they become both consumers and video producers. This media environment has become a serious social problem as it is linked to the production and distribution of fake news videos for various purposes. The emergence of fake news videos is a social problem as they aim to generate political manipulation or profit. There are numerous different kinds of fake news videos.

Among these videos, the new user-generated content (UGC) plays a role in providing news, and in some cases, it has been actually used as a source of news. However, there are various side effects of fake news in that it can cause confusion and misjudgment among the public. Several existing fake news video detection studies rely on the tampering detection algorithm [1]. This method

tries to detect fake news videos by only focusing on the image. However, it is also important to detect whether the information from metadata, as well as the image in the video, is true. Thus, this study presents some ways to use the video's metadata in addition to the visual information to detect fake news videos effectively. In particular, comments as one metadata can play an essential role in detecting fake news videos effectively. As users write them and several users have a critical attitude to the contents of the videos, some comments from a user can provide crucial evidence for fake news videos. In addition, the subjects and domains of fake news videos commonly shift very fast, making fake news video detection difficult. Therefore, we need to develop an efficient model to adapt new subjects and domains easily.

Recently, domain knowledge has been utilized extensively in various natural language processing tasks [2]. We apply a feature selection technique to extract major characteristics that could be used to distinguish fake news videos. The YouTube platform has users from several countries and Internet terms of various expressions. We propose an automatic domain knowledge construction method with various features existing in fake news videos for each subject and domain using feature selection techniques. The proposed method enables the rapid construction of domain knowledge using the feature selection technique based on correlation coefficients. Because the domains of fake news videos commonly

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change very rapidly, it is highly significant to build domain knowledge quickly and automatically whenever domains of fake news videos change. However, since the domain knowledge acquired through the proposed method is extracted from the comments of news videos, there is a weakness in that it is difficult to extract domain features of some fake news videos without enough number of comments. Moreover, various previous studies have effectively detected fake news using second-order user behaviors such as the number of likes, reposts, and shares in the fake news [3,4]. Similarly, in this study, we generated more reliable comments embedding than the existing method using the number of likes of the users' comments. The likes in comments are hereafter referred to as "hit-likes." In other words, domain information is reflected in comments using domain knowledge. The authority of comment embedding is estimated using "hit-likes" to create effective comment embeddings for fake news video detection.

In addition, we attempt to fuse multimodal data using the relationship of metadata extracted from a YouTube video and image information from the thumbnail of the video. Previous studies used texts and images to solve multimodal data tasks by concatenating text and image features at the same weights. However, we figured out how to apply different weights to reflect various aspects of images and text features. In fact, the reflection ratio of text and image features according to the characteristics of the video is significant to detect fake news videos effectively. In particular, previous studies encoded all video frames, which is very inefficient in terms of speed and memory. Herein, we encoded image information with a video thumbnail consisting of several core video frames of YouTube videos. This encoding approach makes our encoding task faster and more effectively than conventional video encoding methods. Moreover, the proposed method also identifies the differences in stance between title/description, and video produced by the video producer and users' comments. In YouTube, this type of stance difference becomes more important for detecting fake news videos because different writers typically generate title/description/video and comments.

We have developed three technologies to build an effective fake news video detection system.

1) Technology that uses the domain knowledge and "hit-likes" of comments to create the comments embedding that is effective in detecting fake news videos; 2) Multi-modal encoding technology for reflecting the importance of the image and text features according to the characteristics of the video; 3) Technology to detect the stance difference between title/description/video and user's comments.

The proposed method was verified through experiments; We achieve an F1-score of 0.93, which is higher than those of other comparison models in all the test datasets. Section 2 explains the details of the proposed model and Section 3 gives the information of datasets used in experiments. In Section 4, we present experiments to verify the contribution of the proposed model in detail. Finally, we provide conclusions and future work in Section 5.

2. Proposed model

Fig. 1 shows the overall configuration diagram of the proposed fake news video detection model (FVDM). Section 1 describes the baseline model, and Sections 2.5 describe the details of the FVDM model.

2.1. BASELINE MODEL

To verify the effectiveness of the proposed model, we employed the latest model to detect fake news videos using the same benchmark dataset as the proposed model: the UcNet (Unified comments Net) as a baseline model. An example of simple features

is the like/dislike ratio and the percentage of violent words in the title. Each video was encoded in simple features, and then its comments were encoded using word2vec. Thereafter, LSTM was used to generate comment embeddings. The comments embedding of the video was obtained by simply averaging the number of comments. Subsequently, the approach detected fake news videos through simple features and concatenated them through a fully connected layer.

2.2. Comments encoding of FVDM

2.2.1. Detailed explanation of domain knowledge

Several video domains exist on YouTube. Moreover, manually building knowledge for such domains is challenging. Therefore, domain knowledge was built using only the training set in this study. In this study, a feature selection technique is applied, and the Pearson correlation coefficient technique is used to select features that could distinguish between fake and real news videos. The correlation coefficients of the features were sorted so that features related to fake news video comments were ranked higher. Subsequently, features were extracted based on a threshold. In our experiments, the built domain knowledge indicated a better performance than the model without domain knowledge at all thresholds. Therefore, the threshold value can be selected based on the speed and performance to build domain knowledge. In this study, all the experiments were conducted by setting the upper 60% threshold, which demonstrated the highest performance based on the experimental results for setting the optimal threshold value. Comment data written in various languages coexist owing to the nature of YouTube. In particular, because the comments are written in several languages, various cultures must be understood to reflect the contents of the languages belonging to the training set. However, this task was extremely difficult. Therefore, we automatically extracted the features of various languages in the training set using a feature selection technique.

2.2.2. Comments encoding using hit-likes

All words are transformed into word pieces using multilingual BERT (Bidirectional Encoder Representations from Transformers) [5] to encode comments in the proposed FVDM. YouTube contains many videos with comments written in diverse languages as well as English. Therefore, the Multilingual BERT is an effective tool to effectively encode comments written in various languages, not just English. $x_k \in \mathbb{R}^d$ is a token vector and d is the dimension of the vector. We define the i th encoded comment as p_i and divide p_i into word piece units to generate comment embedding H^i using Bi-LSTM, where $H^i = [\vec{h}_i; \overleftarrow{h}_i]$. The existing studies have made the comments embedding by simply averaging the embeddings of the total comments. However, we create the comments embedding of the video through the weighted sum of the comments by a specific feature, the number of "hit-likes."

$$C_{\text{hit-likes}} = \{c_1, c_2, \dots, c_N\}, \quad (1)$$

where N is the number of comments of a video, and c_i is the number of "hit-likes" for comment p_i . The sum of the number of "hit-likes" in the total comments is calculated as follows:

$$C_{\text{total}} = \sum_{i=1}^N (c_i + 1). \quad (2)$$

The number one is added to the number of "hit-likes" in Eq. (2) because a comment with zero "hit-likes" must have a weight for the weighted sum. The video comments embedding, H_{comments} , is generated by the weighted sum of comment embed-

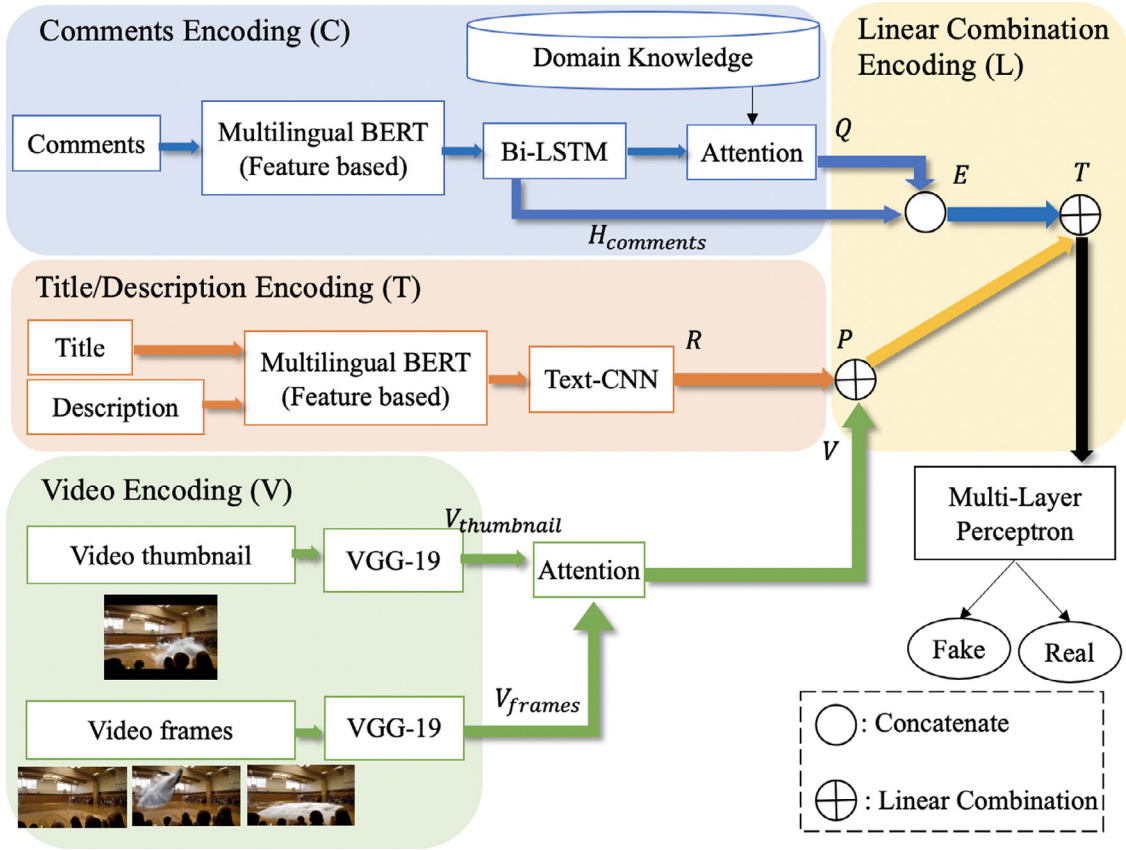


Fig. 1. Architecture of the Fake news Video Detection Model (FVDM).

dings as follows:

$$H_{comments} = \sum_{i=1}^N \frac{c_i + 1}{c_{total} + N} H^i \quad (3)$$

After analyzing the video comments, we thought that they could have authority. Comments with a high number of “hit-likes” can be interpreted as that many other people agree with their thoughts. This method showed better performance than making the video comment embedding by simple averaging of the comment embeddings.

2.2.3. Comments encoding using domain knowledge

The domain knowledge built up in advance is denoted as K . $K \in \mathbb{R}^{w \times d}$, where w is the number of features in the domain knowledge, and the feature of each domain knowledge is expressed as a vector of dimension d . The embedding of a feature included in K is built by averaging the embeddings of the feature through multilingual BERT from all sentences containing the feature in the training set. The attention scores between $H_{comments}$ from Eq. (4) and K are calculated as follows:

$$A_1 = H_{comments} \cdot W_{Attention1} \cdot K, \quad (4)$$

where A_1 is a vector, $A_1 \in \mathbb{R}^w$. Top Y domain knowledge features with high attention scores in A_1 are extracted as additional features for comments. Y is set to 4, which yielded the best performance in the experiment. A set extracted from the top Y feature vectors is denoted as $U = \{b_1, b_2, \dots, b_Y\}$, where $b_i \in \mathbb{R}^d$, and a set of corresponding attention scores for b_i is as $\{a_1, a_2, \dots, a_Y\}$. Subsequently, the extracted features are generated by the weighted sum to create a vector Q as follows:

$$\alpha_i = \text{Softmax}(a_i), \quad (5)$$

$$Q = \sum_{i=1}^Y \alpha_i b_i, \quad (6)$$

where Q is the expanded feature vector containing the potential meaning of the video comments using attention scores in the domain knowledge. Finally, the comments feature E is created by concatenating $H_{comments}$ and Q as in Eq. (7). The symbol $[:]$ means concatenation.

$$E = [H_{comments}; Q], \quad (7)$$

where E is a vector, $E \in \mathbb{R}^{2d}$. The feature embedding dimension of comments, title/description, and video features is the same as $\mathbb{R}^{2d}$ for the linear combination of them.

2.3. Title/description encoding of FVDM

The CNN (Convolutional Neural Network) is used to extract topic information from title/description of video [6]. The sentences are transformed into word piece units using multilingual BERT similarly as in comments encoding. Assuming that n tokens exist in title/description, the title/description vector is constructed as follows:

$$x_{1:n} = [x_1; x_2; \dots; x_n], \quad (8)$$

where the symbol $[:]$ implies concatenation. In general, $x_{k:k+j}$ is expressed as $x_k, x_{k+1}, \dots, x_{k+j}$. The CNN filter is defined as $W_{CNN} \in \mathbb{R}^{\beta \times d}$. β is the filter size.

$$u_k = \text{ReLU}(W_{CNN} \cdot x_{k:k+\beta-1} + b), \quad (9)$$

where ReLU (Rectified Linear Unit) is a nonlinear function. If the filter is applied to all n tokens, then u_k can be expressed as a fea-

ture map G as follows:

$$G = [u_1, u_2, \dots, u_{n-\beta+1}]. \quad (10)$$

Feature map G selects the maximum value z through the max-over-time pooling function. This process is repeated by the number of filters γ . Finally, by concatenating all the γ maximum values, title/description is encoded as $Z_{1:\gamma} \in \mathbb{R}^\gamma$. $Z_{1:\gamma}$ is the vector representation of the holding topic information from the title/description. The feature vector, $Z_{1:\gamma}$, with the topic information of title/description is projected as shown in the following Eq. (11) so that the dimension of the feature vector becomes $\mathbb{R}^{2d}$.

$$R = \tanh(W_\alpha \cdot Z_{1:\gamma}), \quad (11)$$

where R is a vector, $R \in \mathbb{R}^{2d}$.

2.4. Video encoding of FVDM

Unlike the existing video encoding methods, we encode the thumbnail and parts of the video frames to accelerate encoding time and obtain effective encoding results. The pre-trained VGG-19 (Visual Geometry Group-19) is employed to efficiently extract the visual features from the thumbnail and video frames of the video individually[7]. To fairly evaluate the contribution of the proposed method, we used VGG-19 used in the comparison models. A fully connected layer is added to the last layer of the VGG-19 network to create the final visual representation for the thumbnail. The visual feature representation of video thumbnail is denoted as $V_{thumbnail} \in \mathbb{R}^{2d}$, and the visual representation of i th video frames is as $V_{frame(i)} \in \mathbb{R}^{2d}$. We extract J frames most similar to $V_{thumbnail}$ by using cosine similarity scores between $V_{thumbnail}$ and $V_{frame(i)}$ as follows:

$$\cos \theta = \frac{V_{thumbnail} \cdot V_{frame(i)}}{\|V_{thumbnail}\| \|V_{frame(i)}\|}. \quad (12)$$

The set, L , is composed of $V_{thumbnail}$ and the encoding vectors of extracted J frames. The reason of adding $V_{thumbnail}$ to the L set is that some thumbnails are not from the original video frames.

$$L = \{V_{frame(1)}, V_{frame(2)}, \dots, V_{frame(J)}, V_{frame(J+1)}\}. \quad (13)$$

$V_{thumbnail}$ and the set L , are used to generate the vector of the attention scores, A_2 , as follows:

$$B_1 = V_{thumbnail} \cdot W_{Attention2} \cdot L, \quad (14)$$

where $B_1 = [a_1, a_2, \dots, a_j, a_{j+1}]$. The vector V could be created through the weighted sum method as follows:

$$\beta_i = \text{Softmax}(a_i), \quad (15)$$

$$V = \sum_{i=1}^{J+1} \beta_i V_i, \quad (16)$$

where V is a vector, $V \in \mathbb{R}^{2d}$, containing the core visual information from video thumbnail and important video frames.

2.5. Linear combination encoding of FVDM

In previous studies, linear combinations have been used to effectively fuse the two feature vectors from text and video[8]. In our case, we combine three feature vectors (V as a video encoding vector, R as a title/description vector, and E as a comment vector) by the following processes. First, to learn the relationship between the video and the title/description features, m_1 is adjusted to reflect the relationship between the video and the title/description

Table 1

YouTube video dataset specifications.

Dataset	FVCv2.0	FVCv3.0	VAVD	MYVC	Full dataset
Fake	96	1212	117	142	1411
Real	72	1173	331	145	1501

features to learn the encoding ratio for fake news video detection effectively. m_1 can be calculated as follows:

$$m_1 = \sigma(w_\beta \cdot [V; R]), \quad (17)$$

where m_1 is a scalar value. By the linear combination of V and R , a vector, P , is generated by Eq. (18).

$$P = m_1 \cdot V + (1 - m_1) \cdot R, \quad (18)$$

where P is a feature vector containing the visual information of the video and the topic information of the title/description. Moreover, the same process is applied to concatenate P and E that include video's comments information. m_2 in Equation (19) learns the stance between the video and the title/description created by the video producer and the users' comments.

$$m_2 = \sigma(w_\gamma \cdot [E; P]), \quad (19)$$

$$T = m_2 \cdot E + (1 - m_2) \cdot P, \quad (20)$$

where m_2 is a scalar value.

Finally, T contains the total information from comments, title/description, and video information according to the following Eq. (21).

$$T = m_2 \cdot E + (1 - m_2) \cdot m_1 \cdot V + (1 - m_2) \cdot (1 - m_1) \cdot R. \quad (21)$$

The vector T is used as an input for the MLP layer to effectively detect fake news videos. The detailed steps of the proposed FVDM (Fake news Video Detection Model) are summarized in Algorithm 1.

3. Datasets

3.1. Youtube datasets

Table 1 shows the datasets used in the experiments, which were conducted using five datasets. FVC2.0 and FVC3.0 were developed in the InVID (In VIDEO veritas) project [9]. The FVC dataset is composed of fake and real news videos of various topics, such as politics, sports, and accidents. Next, VAVD (Volunteer Annotated Video Dataset) is a publicly available video dataset developed in [10], and MYVC (Misleading Youtube Video Corpus) is a dataset we collected. First, fake and real news were collected from popular fact-checking websites, such as Snopes, Fact Check-CNN, PolitiFact, and then fake news video data were collected by searching in YouTube for these collected fake and real news. Two researchers evaluated the collected video data. Cohen's Kappa coefficient of 0.92 was obtained; thus, the collected video data can be regarded as reliable data. Finally, we created a full dataset by combining all four datasets, and duplicate videos were removed from the entire dataset.

4. Experiments

We compared and analyzed the results of this study with UcNet, the latest model that used the same benchmark dataset as our model to determine the contribution of the proposed model. The proposed model is denoted as $FVDM_{T+C+V+L}$ with "Title/Description" (T), "Comment" (C), "Video" (V), and Linear combination (L) encodings.

Algorithm 1 Pseudo-Code of proposed fake news video detection model.

Input: Data $D = \{d_1, \dots, d_m\}$, $d_i = \{C, T\&D, M\}$

- 1: # C is comments.
- 2: # $T\&D$ is title and description.
- 3: # M is thumbnail and frames.
- 4: **for** d_i in the given Data $D = \{d_1, d_2, \dots, d_m\}$
- 5: $E = \text{Generate Comments Embedding}(C)$
- 6: $R = \text{Generate Title\&Description Embedding}(T\&D)$
- 7: $V = \text{Generate Video Embedding}(M)$
- 8: $P = \text{Execute linear combination}(V, R)$
- 9: $T = \text{Execute linear combination}(E, P)$
- 10: $\hat{d}_{fake} = \text{Detect fake news video}(T)$
- 11: **Update parameters**($\hat{d}_{fake}$)
- 12:
- 13: **Generate Comments Embedding**(C)
- 14: **for** p_i in the given comments $C = \{p_1, p_2, \dots, p_m\}$
- 15: $H^i = \text{Bi-LSTM}(p_i)$ # p_i is i th comment.
- 16: $H_{comments} = \sum_{i=1}^N \frac{c_i+1}{c_{total}+N} H^i$
- 17: $A_1 = H_{comments} \cdot W_{Attention1} \cdot K$ # K is domain knowledge.
- 18: **Extract** features with high attention scores from A_1
- 19: $\alpha_i = \text{Softmax}(a_i)$
- 20: $Q = \sum_{i=1}^Y \alpha_i b_i$
- 21: $E = \{H_{comments}; Q\}$
- 22: **Return** E
- 23: **Generate Title\&Description Embedding**($T\&D$)
- 24: $R = \text{CNN}(T\&D)$
- 25: **Return** R
- 26: **Generate Video Embedding**(M)
- 27: $V_{thumbnail}, V_{frame(i)} = \text{VGG-19}(M)$
- 28: **Extract** frames with high cosine similarity scores between $V_{thumbnail}$ and $V_{frame(i)}$
- 29: $L = \{V_{frame(1)}, V_{frame(2)}, \dots, V_{frame(j)}, V_{frame(j+1)}\}$
- 30: $B_1 = V_{thumbnail} \cdot W_{Attention2} \cdot L$ # B_1 is $[a_1, a_2, \dots, a_j, a_{j+1}]$
- 31: $\beta_i = \text{Softmax}(a_i)$
- 32: $V = \sum_{i=1}^{j+1} \beta_i V_i$
- 33: **Return** V
- 34: **Execute linear combination**(A, B)
- 35: $m_1 = \sigma(w_\beta \cdot [A; B])$
- 36: $C = m_1 \cdot A + (1 - m_1) \cdot B$
- 37: **Return** C
- 38: **Detect fake news video**(T)
- 39: $\hat{d}_{fake} = \text{MLP}(T)$ # $\hat{d}_{fake}$ is the probability of news video being a fake one.
- 40: **Update parameters**($\hat{d}_{fake}$)
- 41: $\mathcal{L} = -\mathbb{E}_{(vc,y) \sim (vc,Y)} [y \log(\hat{d}_{fake}) + (1-y) \log(1 - \hat{d}_{fake})]$
- 42: # Y is the label set that indicates whether the video is fake, and VC is the set of all news videos.
- 43: $\theta_f = \theta_f - \lambda \frac{\partial \mathcal{L}}{\partial \theta_f}$

4.1. Implementation details

The experiments were implemented using Pytorch under Ubuntu 16.04 LTS with TITAN X GPU. The training time of our proposed model is 0.17 sec/video, and the execution time is 0.1 sec/video. The proposed model was trained using Adam with a learning rate of 0.0001[11]. Also, the time complexity of our proposed model is $n^2 \cdot d + 3 \cdot n \cdot d^2$, where n is the sequence length and d is the dimension. The final computational complexity of the proposed model will be $O(n^2)$.

4.2. Effect of comments authority

Fig. 2 shows the F1-score results of the two encoding methods, average and weighted sum to reflect comment authority, for comments in all datasets using $FVDM_{T+C+V+L}$. The weighted sum method creates comments embedding as shown in Eq. (3), and the average method generates comments embedding by adding up all the comment embeddings included in one video and dividing it by the number of comments. As a result, the proposed method outperforms the existing embedding method; simple average and the hypothesis that the number of “hit-likes” in comments can reflect the comments authority.

4.3. Effect of video encoding

Fig. 3 shows F1-score performance changes according to the rate of video frames in the video. The frame ‘0’ in Fig. 3 implies the case where a video thumbnail is only used. We encoded only top 10% frames based on the similarity between the video thumbnail and the video frames. The experimental results showed the highest performance when encoding 10% frames of each video. It can be inferred that increasing the encoding ratio degrades performance for the following reasons: First, only about 10% frames of each video have effective features to detect fake news videos. Second, a large ratio of video frames generates noise when detecting fake news videos. The results show that the proposed method, including parts of the video frames, can detect fake news videos faster and more effectively than conventional methods, which is encoding all frames of the video.

4.4. Effect of linear combination encoding

Fig. 4 shows the encoding parameter ratio of comments, title/description, and video according to the concatenation method and video type. The linear combination method creates a vector T by linear combination as shown in Eq. (21), and the concatenation method generates a vector T by concatenating three vectors, vector E , vector V , and vector R in Eq. (21). (a) and (c) present the different encoding parameter ratios for fake news video thumbnails. In the case of (a), video has a strong clue to detect a fake news video; the thumbnail of (a) has a fake point, “pink smoke.” Conversely, the case of (c) does not add any clue for fake news video detection on video itself. Thus, the former case has a high parameter weight for video, while the latter case has a high parameter weight for comments. The case of (b) for real news video can be contrasted to the case (a) and (c) for fake news video. In that case, the parameter weight of title/description has the highest weight because a creator can naturally upload the title/description for their video. We conducted several experiments using the attention mechanism. For example, other comment embedding P was generated by calculating the attention scores between video embedding V and H ; H is $\{H^1, \dots, H^N\}$, H^i is the embedding vector of each comment, and N is the number of comments. Then, domain knowledge embedding Q , comment embedding P , and Title/description embedding R are concatenated as an input to the fake news classifier. As a result, the F1-score was 80.22%, which showed lower performance than one of our proposed model.

Fig. 5 shows the average encoding rate of comments, title/description, and video in the entire dataset. The average encoding weight of comments is 44.33% and 10.2% for fake and real news videos, respectively. In contrast, title/description has a much higher encoding rate in the real than in the fake news video. The experimental results show that our linear combination method encodes three features effectively and efficiently depending on the characteristics of the video and prevents learning by relying on video-specific events to prohibit over-fitting of the model.

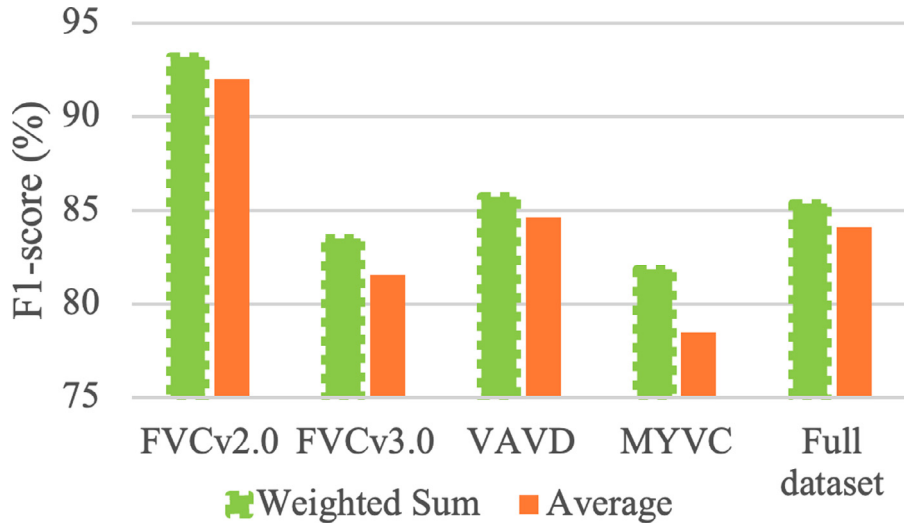


Fig. 2. The Impact of comments authority on fake news video detection.

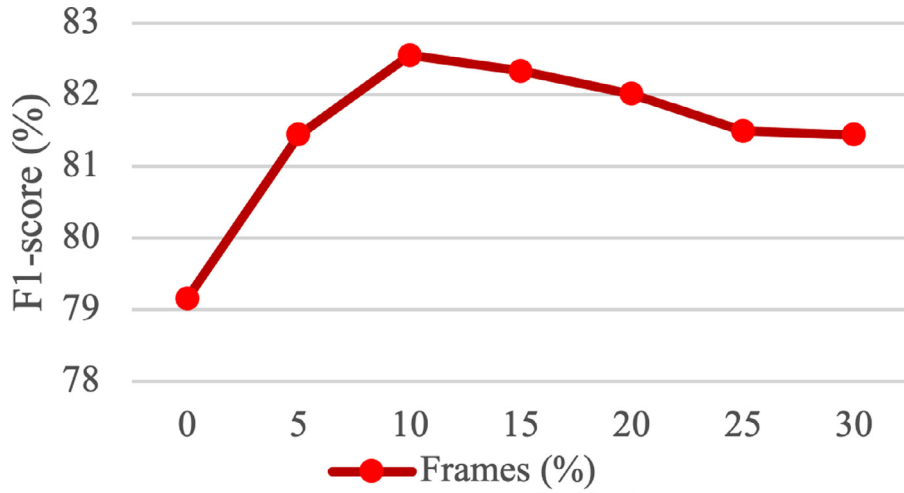


Fig. 3. Performance variation of F1-scores according to the number of frames of the video.

Table 2

Overall F1-score comparison (%) among models on the different datasets.

Models	FVCv2.0	FVCv3.0	VAVD	MYVC	Full dataset	Average	VAVD (AUC)
UcNet [10]	82.61	73.05	75.22	70.66	72.16	74.74	0.81
MVAE [12]	89.54	78.11	80.88	80.66	81.04	82.04	0.84
SpotFake [13]	92.77	82.54	83.34	81.25	82.01	84.38	0.86
$FVDM_{T+C+V}$ w/ word2vec	86.67	77.91	78.15	73.21	76.18	78.42	0.82
$FVDM_{T+C+V}$ w/ word2vec w/ domain knowledge	89.76	80.56	82.33	77.23	79.41	81.85	0.84
$FVDM_{T+C+V+L}$ w/ word2vec w/ domain knowledge	91.82	82.91	84.84	80.77	80.97	84.26	0.87
$FVDM_{T+C+V+L}$ w/ multilingual BERT w/ domain knowledge	93.21	83.50	85.71	81.82	82.55	85.34	0.88

4.5. Comparison with existing systems

Table 2 shows the F1-scores obtained with the $FVDM_{T+C+V+L}$, MVAE [12], SpotFake [13], and UcNet [10]. For accurate performance comparison, we conducted the experiments in the same experimental environment as a conventional fake news video system [10]. As in the study by [10], the VAVD data were divided into the train set and test set in 7:3 ratio to evaluate the models, and for all other datasets, 5-fold cross-validation was used to perform the evaluation. We evaluated the model with 5-fold cross-validation and did the t -test between our proposed model and SpotFake on

the full dataset. As a result of the t -test, the p -value was 0.003. Since the p -value is less than 0.01, our experimental results are strongly significant. UcNet is the only study that detects fake news videos based on my best knowledge. Thus, two fake news detection systems, MVAE and SpotFake, using news images and texts were selected to be compared with the proposed method. In our experiments, we trained them using the same video V , title/description R , and comments embedding $H_{comments}$ of the proposed model. From Table 2, it can be seen that $FVDM_{T+C+V+L}$ outperforms the existing models on all datasets. In addition, the reason why we encoded title/description with CNN in our proposed model is to extract topic



Fig. 4. “Comments,” “Title/Description,” and “Video” encoding ratio according to the combination method and video type.

information from the video. When using CNN to encode the title/description, the F1 score based on the full dataset was 82.55%, but when encoding with LSTM, the F1 score was 81.01%. These results demonstrate that encoding the topic information of a video using the title/description with CNN is effective in detecting fake news videos.

4.6. Effect of domain knowledge

The results listed in Table 2 show that the difference in performance between the addition of domain knowledge and the absence of domain knowledge is 3.43%p on average, and this experimental result proves that domain knowledge is very effective in detecting fake news videos. This indicates that constructing the domain knowledge for each dataset is efficient. For example, the domain knowledge of the FVCv3.0 dataset contained fea-

tures such as “deepfake,” “cgi,” and “photoshop.” In contrast, in the MYVC dataset, the videos were not manipulated; however, the dataset contains numerous videos with false information. The domain knowledge of the MYVC dataset contained features such as “fakenews” and “factcheck.” As the features differed based on the domain type of the fake news video, their performances can be improved further if each dataset is used to construct its domain knowledge.

4.7. Threshold selection of domain knowledge construction

Fig. 6 shows the F1-score performance changes based on the Top K% extraction when constructing the domain knowledge by feature selection. As all K% indicates higher performances than those without domain knowledge, the usefulness of domain knowledge was proven. Consequently, the domain knowledge for

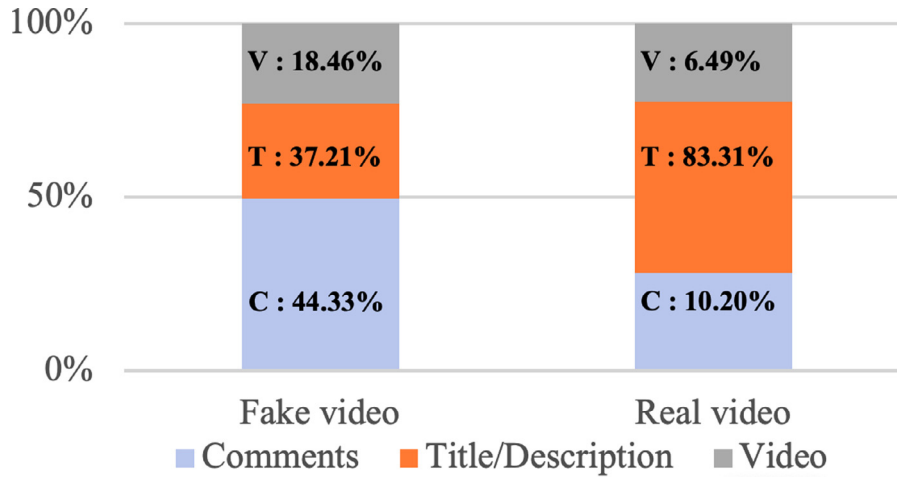


Fig. 5. Encoding ratio of “Comments,” “Title/Description,” and “Video” on the full dataset for fake and real data.

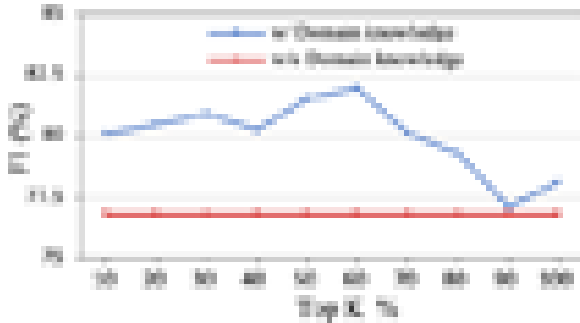


Fig. 6. Comparison of F1-scores on the full dataset following Top K % extracted from the feature selection.

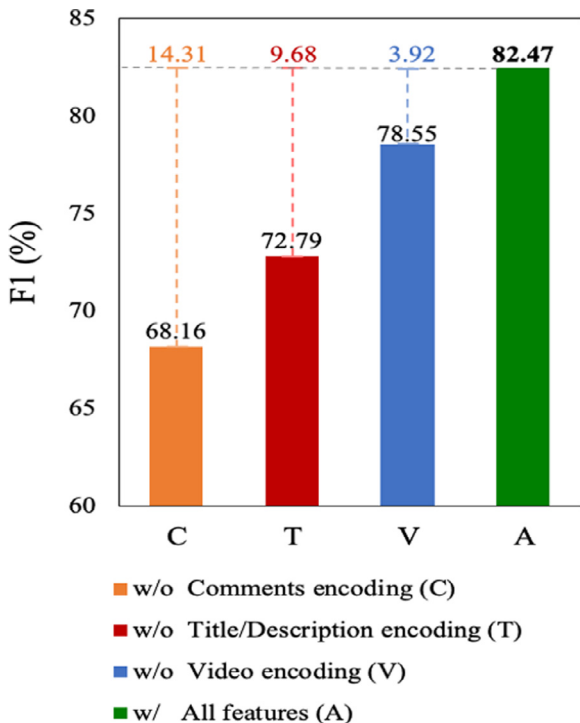


Fig. 7. Ablation tests of $FVDM_{T+C+V}$ on the full dataset.

each data set was constructed by extracting the top 60% performance.

4.8. Ablation study

Fig. 7 shows the results of the contribution assessment for each factor. To analyze the contribution of three variables, “Title/Description,” “Comments,” and “Video” used in this experiment, we removed each encoding method in Fig. 1 one by one to identify the difference in the contribution of the factors. Among the three features we used in the experiment, the most effective feature for detecting fake news videos was comments. Because comments is the metadata created by the users, it contains key features for detecting fake news videos. In addition, we assigned authority to comments, making the information in the comments more reliable. However, the “Title/Description” and “Video” does not contain relatively significant features for identifying fake news videos as it was created by video creators to trick users. Moreover, the lowest performance contribution of video factors is because the video data does not contain some video clips that people can distinguish whether they were real or fake by itself. It implies that the level of video production by fake news video creators has recently become so high that ordinary people can find it difficult to determine the difference between real and fake just by watching the video. This study can provide evidence that it is much more effective to detect fake news videos using all factors such as “Title/Description,” “Comments,” and “Video.”

5. Conclusion and future work

This study has evaluated a new deep-learning-based fake news video detection model. We used domain knowledge to perform learning by reflecting the potential meaning of comments, which helped us detect fake news videos and learned robustly without relying on domains. In addition, the feature selection method has greatly contributed to detecting fake news videos by extracting features from various languages. We used the linear combination to efficiently adjust the encoding rate for each characteristic of the video and efficiently detect fake news videos. Compared with the performance of SpotFake, the latest model for detecting fake news videos. We achieve an F1-score of 0.93, which is higher than those of other comparison models in all the test datasets. Our follow-up study will focus on using subtitles and voice in the video to detect fake news videos more effectively. In addition, we plan to utilize domain agnostic features to create a model that effectively detects fake news videos even though the fake news domains change fast.

Confirmation of Authorship

1. This manuscript, or a large part of it, has not been published, was not, and is not being submitted to any other journal.

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5. All authors each made a significant contribution to the research reported and have read and approved the submitted manuscript.

Declaration of Competing Interest

The researcher claims no conflicts of interest.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.patrec.2022.01.007](https://doi.org/10.1016/j.patrec.2022.01.007).

References

- [1] R.C. Pandey, S.K. Singh, K.K. Shukla, Passive forensics in image and video using noise features: a review, *Digital Invest.* 19 (2016) 1–28.
- [2] N. Locascio, K. Narasimhan, E. DeLeon, N. Kushman, R. Barzilay, Neural generation of regular expressions from natural language with minimal domain knowledge, *arXiv preprint arXiv:1608.03000* (2016).
- [3] J.C. Reis, A. Correia, F. Murai, A. Veloso, F. Benevenuto, Explainable machine learning for fake news detection, in: *Proceedings of the 10th ACM conference on web science*, 2019, pp. 17–26.
- [4] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, H. Liu, Fakenewsnet: a data repository with news content, social context, and spatiotemporal information for studying fake news on social media, *Big Data* 8 (3) (2020) 171–188.
- [5] T. Pires, E. Schlinger, D. Garrette, How multilingual is multilingual bert? *arXiv preprint arXiv:1906.01502* (2019).
- [6] Y. Kim, Convolutional neural networks for sentence classification, *arXiv preprint arXiv:1408.5882* (2014).
- [7] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, *arXiv preprint arXiv:1409.1556* (2014).
- [8] J.-H. Kim, S.-W. Lee, D.-H. Kwak, M.-O. Heo, J. Kim, J.-W. Ha, B.-T. Zhang, Multimodal residual learning for visual qa, *arXiv preprint arXiv:1606.01455* (2016).
- [9] O. Papadopoulos, M. Zampoglou, S. Papadopoulos, Y. Kompatsiaris, D. Teyssou, Invid fake video corpus v2. 0 (version 2.0), Dataset on Zenodo (2018).
- [10] P. Palod, A. Patwari, S. Bahety, S. Bagchi, P. Goyal, Misleading metadata detection on youtube, in: *European Conference on Information Retrieval*, Springer, 2019, pp. 140–147.
- [11] D.P. Kingma, J. Ba, Adam: a method for stochastic optimization, *arXiv preprint arXiv:1412.6980* (2014).
- [12] D. Khattar, J.S. Goud, M. Gupta, V. Varma, Mvae: Multimodal variational autoencoder for fake news detection, in: *The World Wide Web Conference*, 2019, pp. 2915–2921.
- [13] S. Singhal, R.R. Shah, T. Chakraborty, P. Kumaraguru, S. Satoh, Spotfake: A multi-modal framework for fake news detection, in: *2019 IEEE Fifth International Conference on Multimedia Big Data (BigMM)*, IEEE, 2019, pp. 39–47.